Cold Storage module — actual effort by API

Summary for sub modules

Sr. No.	Sub-module	Endpoints	Dev Effort (in hours)
1	Cold storage facility master	9	18.0
2	Block and rack configuration	9	17.0
3	Crop receipt and intake	12	17.0
4	Shelf-life master	11	14.0
5	Storage rate master	9	14.0
6	Crop status monitoring and alerts	9	16.0
7	Crop return and payment	13	18.0
8	Requirement analysis, schema design, EJS template, regression suite, defects remediation, misc. etc		76.0
	Total	**72**	**190.0**

Additional effort for change in requirement for sub modules

Sr. No.	Sub-module	Endpoints impacted	Dev Effort (in hours)
2	Block and rack configuration	4	2
3	Crop receipt and intake	8	7.5
5	Storage rate master	5	2.5
6	Crop status monitoring and alerts	3	1.5
7	Crop return and payment	5	5
8	Requirement analysis, schema design, EJS template, regression suite, defects remediation, misc. etc		8.5
	Total	**25**	**27**

	Grand Total (Dev Effort in hours)		**217.0**

Cold storage facility master

Base path /cold-storage · 9 endpoints

Sr. No.	Method	Endpoint	Dev Effort (in hours)
1	GET	/types	3.0
2	POST	/	4.0
3	GET	/	2.0
4	GET	/code/:cs_id	2.0
5	GET	/:id	1.0
6	PATCH	/restore/:id	2.0

7	PATCH	/:id	2.0
8	DELETE	/permanent/:id	1.0
9	DELETE	/:id	1.0
	Total		**18.0**

Block and rack configuration

Base path /cold-storage/:cs_id/block-config · 9 endpoints

Sr. No.	Method	Endpoint	Dev Effort (in hours)
1	POST	/	4.0
2	PATCH	/ (bulk upsert)	3.0
3	GET	/	2.0
4	GET	/type/:storage_t	2.0
5	GET	/:id	1.5
6	PATCH	/restore/:id	1.5
7	PATCH	/:id	1.0
8	DELETE	/permanent/:id	1.0
9	DELETE	/:id	1.0
	Total		**17.0**

Crop receipt and intake

Base path /cold-storage/:cs_id/consignment · 12 endpoints

Sr. No.	Method	Endpoint	Dev Effort (in hours)
1	POST	/	3.0
2	GET	/	2.0
3	GET	/racks	2.0
4	GET	/racks/available	1.0
5	GET	/code/:consignm	1.0
6	GET	/print/receipt/:id	2.0
7	GET	/receipt/:consign	1.0
8	GET	/:id	1.0
9	PATCH	/restore/:id	1.0
10	PATCH	/:id	1.0
11	DELETE	/permanent/:id	1.0
12	DELETE	/:id	1.0
	Total		**17.0**

Shelf-life master

Base path /cold-storage-shelf-life · 11 endpoints

Sr. No.	Method	Endpoint	Dev Effort (in hours)
1	POST	/	2.0
2	PATCH	/ (bulk upsert)	2.0
3	GET	/	1.0
4	GET	/matrix	2.0
5	GET	/resolve	1.0
6	GET	/crop/:crop_id	1.0
7	GET	/:id	1.0
8	PATCH	/restore/:id	1.0
9	PATCH	/:id	1.0
10	DELETE	/permanent/:id	1.0
11	DELETE	/:id	1.0
	Total		**14.0**

Storage rate master

Base path /cold-storage-rate · 9 endpoints

Sr. No.	Method	Endpoint	Dev Effort (in hours)
1	POST	/	3.0
2	GET	/	1.5
3	GET	/current	2.0
4	GET	/resolve	1.5
5	GET	/:id	1.0
6	PATCH	/restore/:id	1.0
7	PATCH	/:id	2.0
8	DELETE	/permanent/:id	1.0
9	DELETE	/:id	1.0
	Total		**14.0**

Crop status monitoring and alerts

Base path /cold-storage-alert · 9 endpoints

Sr. No.	Method	Endpoint	Dev Effort (in hours)
1	GET	/dashboard	2.5
2	GET	/cards	2.5
3	GET	/log	1.5
4	GET	/log/:id	1.0

5	GET		/export	1.0
6	GET		/consignment/:c	1.0
7	POST		/reclassify	2.5
8	POST		/dispatch	2.5
9	POST		/notify/:consignr	1.5
	Total			**16.0**

Crop return and payment

Base path /cold-storage/:cs_id/return · 13 endpoints

Sr. No.	Method	Endpoint	Dev Effort (in hours)
1	GET	/preview/:consig	2.5
2	POST	/	3.5
3	GET	/	1.5
4	GET	/code/:payment	0.5
5	GET	/consignment/:c	0.5
6	GET	/print/slip/:id	2.0
7	GET	/print/receipt/:id	2.0
8	GET	/slip/:payment_r	1.0
9	GET	/:id	0.5
10	PATCH	/restore/:id	1.0
11	PATCH	/:id	1.0
12	DELETE	/permanent/:id	0.5
13	DELETE	/:id	1.5
	Total		**18.0**

Requirement analysis, schema design, EJS template, regression suite, defects remediation, misc. etc

Sr. No.	Activity	Dev Effort (in hours)
1	Requirement analysis and domain modelling	12.0
2	Database schema design	10.0
3	Printable documents (EJS templates, QR)	8.0
4	Regression test suite (7 Postman collections)	20.0
5	Defect investigation, fixing and testing	8.0
6	Shared code utilities	4.0
7	Miscellaneous	10.0
8	SMS integration	4.0
	Total	**76.0**

Additional effort for change in requirement for sub modules

Block and rack configuration

Base path /cold-storage/:cs_id/block-config — 4 endpoint(s) impacted

	Method	Endpoint	Dev Effort (in hours)	What changed
1	POST	/	0.5	lock_in_days added to the payload, minimum 15 (one billing block)
2	PATCH	/ (bulk upsert)	0.5	lock_in_days on each of the four storage types
3	PATCH	/:id	0.5	lock_in_days editable, never reprices produce already in store
4	GET	/	0.5	lock_in_days in the response and as a sort field
	Total		**2**	

Crop receipt and intake

Base path /cold-storage/:cs_id/consignment — 8 endpoint(s) impacted

	Method	Endpoint	Dev Effort (in hours)	What changed
1	POST	/	2	number_of_boxes / weight_per_box replaced by units[], each line priced from its own unit-type rate and snapshotted, daily_basket_rate, lock_in_days and estimated_rent derived
2	**POST**	**/:id/advance (r**	2	unit-type rate and snapshotted, daily_basket_rate, lock_in_days and estimated_rent derived
3	**GET**	**/print/advance-**	1	New. Advance receipt PDF (new EJS template); 404 when no advance was collected
4	PATCH	/:id	0.5	units replaced as a set and re-priced; storage-type change re-derives lock_in_days; refused once an advance or any return row exists
5	GET	/print/receipt/:id	0.5	Storage receipt PDF gained the unit lines and charge basis
6	GET	/	0.5	unit fields in the response; lock_in_days and estimated_rent as sort fields
7	GET	/:id	0.5	unit_summary, advance_paid and advance_balance added to the response
8	GET	/code/:consignm	0.5	Same response shape as above
	Total		**7.5**	

Storage rate master

Base path /cold-storage-rate — 5 endpoint(s) impacted

	Method	Endpoint	Dev Effort (in hours)	What changed
1	POST	/	0.5	unit_type added to the key, non-overlap now per (storage type, unit type)
2	GET	/	0.5	unit_type filter and sort; rate_per_box_per_day renamed rate_per_unit_per_day
3	GET	/current	0.5	unit_type filter and sort; rate_per_box_per_day renamed rate_per_unit_per_day

4	GET	/resolve	0.5	unit_type now required
5	DELETE	/:id	0.5	Dependency check narrowed to consignments carrying that unit type
	Total		**2.5**	

Crop status monitoring and alerts

Base path /cold-storage-alert — 3 endpoint(s) impacted

	Method	Endpoint	Dev Effort (in hours)	What changed
1	GET	/dashboard	0.5	total_boxes renamed total_units
2	GET	/cards	0.5	Unit wording in the card and in meta.totals
3	GET	/export	0.5	CSV column header Boxes renamed Units
	Total		**1.5**	

Crop return and payment

Base path /cold-storage/:cs_id/return — 5 endpoint(s) impacted

1	GET	/preview/:consig	1.5	Computation block gained lock_in_days, extra_days, blocks_charged, block_days, days_charged and advance_adjusted
2	POST	/	1.5	Charges read the intake snapshot instead of resolving the rate master backwards; advance set off rather than re-charged; amount_paid validated against the balance, not the total
3	GET	/print/slip/:id	0.5	Return slip prints the unit lines and the day breakdown
4	GET	/print/receipt/:id	1	Payment receipt prints the unit lines and the advance deduction
5	GET	/	0.5	meta.totals gained days_charged and advance_adjusted
	Total		**5**	

Requirement analysis, schema design, EJS template, regression suite, defects remediation, misc. etc

	Activity			Detail
1	Requirement analysis and domain modelling		3	Per-unit pricing, lock-in as a storage-type tariff, 15-day block billing, advance set-off rules
2	Database schema changes		1	cold_storage_consignment_unit table; lock_in_days, daily_basket_rate, estimated_rent and the advance columns; unit_type on the rate master
3	Printable documents (EJS templates)		1.5	Storage receipt, return slip and payment receipt updated; advance receipt template added
4	Regression test suite (5 Postman collections)		1	Consignment, Return, Rate, Block Config and Alert collections reworked for the new payloads
5	Defect investigation, fixing and testing		1	District-wide sequence collision; UTC/local date defects in expiry and alert classification
6	Shared code utilities		1	local-date helpers, rupees-in-words, district-scoped sequence generator
	Total		**8.5**	